

Exam 3 Topics

- Stability → Requires AC analysis background.

- AC Noise

- Non-Linear Circuits

→ Saturation Circuits

→ Circuits w/ Diodes

→ Oscillators

} Not Reviewing
Today!

→ [At least] one topic from Exam 1 & 2

Email/webex me if interested in taking exam on reading days.

Presentations Monday. Plan on 5-10 minutes

Stability & Compensation.

Ability to calculate $T(jf) = a(jf)\beta(jf)$

$$\beta(jf) = -\frac{v_o(jf)}{v_i}$$

Determine Stability from
either $|T(jf_x)| = 1 \rightarrow \phi_m$

or $\angle T(jf_{-180}) = -180 \rightarrow GM$

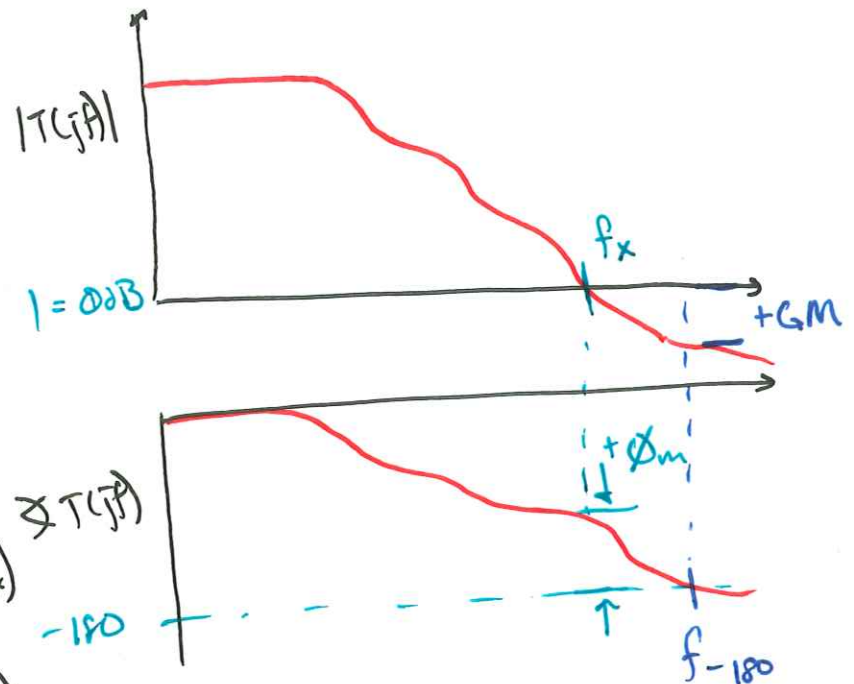
$$\begin{aligned}\phi_m &= \angle T(jf_x) = 180^\circ + a(jf_x) + \beta(jf_x) \\ &= 180^\circ + a(jf_x) - 1/\rho(jf_x)\end{aligned}$$

$$GM = -20 \log(|T(jf_{-180})|)$$

finding f_x : $|T(jf_x)| = |a(jf_x)|/|\beta(jf_x)| \Rightarrow 1 \rightarrow |a(jf_x)| = 1/|\beta(jf_x)|$

~~f_{-180} & $\angle T(jf_{-180})$~~

in finding f_x , f_{-180} , be aware of zero & pole locations
for simplification!



for stability,

$$\phi_m > 0^\circ$$

$$GM > 0 \text{ dB}$$

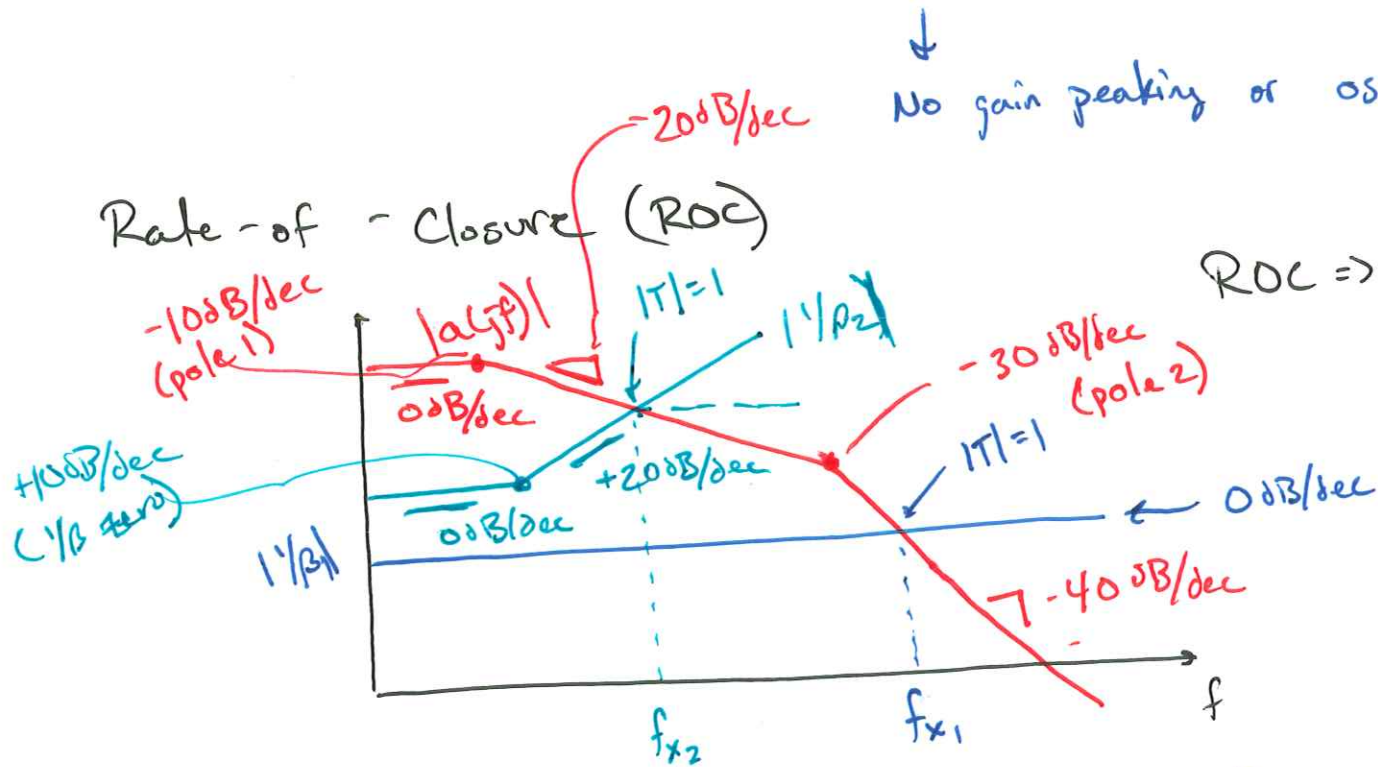
desired:

$$\phi_m > 45^\circ$$

$$GM > 10 \text{ dB}$$



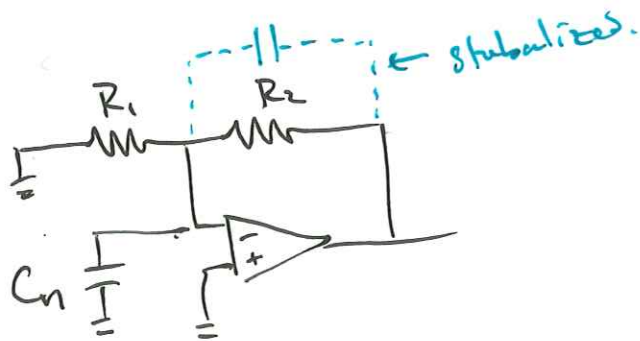
No gain peaking or oscillations.



$$ROC \Rightarrow \phi_m \approx 180^\circ + 4.5^\circ \cdot \text{slope of } |a(jf_x)| - 4.5^\circ \cdot \text{slope of } 1/B(f_x)$$

Ex1: $\phi_m \approx 180 + 4.5(-40) - 4.5(0) = 0^\circ$

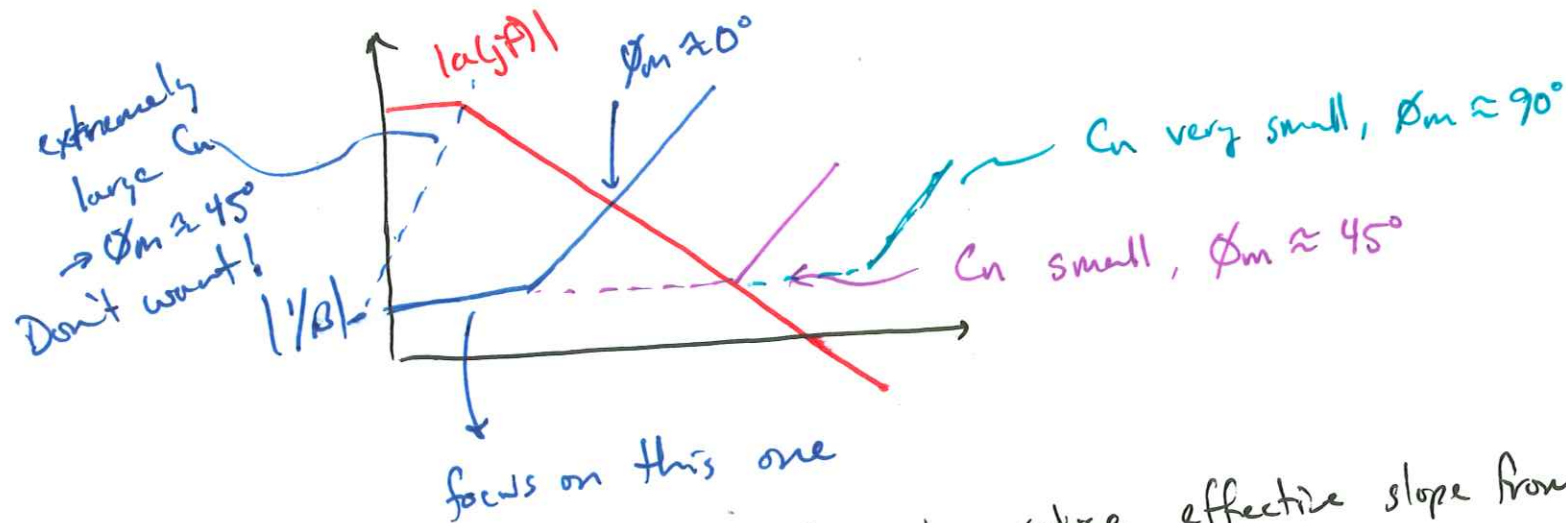
Ex2: $\phi_m \approx 180^\circ + 4.5(-20) - 4.5(-20) = 0^\circ$



$$\beta = \frac{R_1 \parallel Z_{Cn}}{R_1 \parallel Z_{Cn} + R_2}$$

→ Results in a pole in β
zero in $1/\beta$

if $a(jf)$ is GBWP



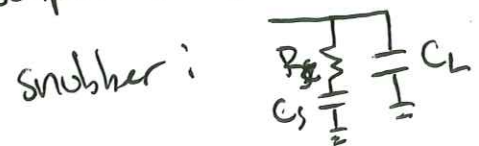
→ Add pole to $1/\beta$ at f_x to reduce effective slope from -20 to -10
→ $\Phi_m \approx 45^\circ$

How - feedback cap. (see above)

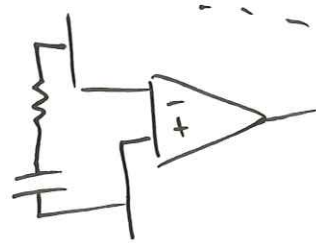
Can also have ~~β~~ added f_p (beta pole) very near f_z (beta zero)
such that they cancel out! → $\Phi_m \approx 90^\circ$

→ At the expense of f_B

Similar behavior & considerations for a C_L (output load)



~~For~~ input lag compensation:



Noise

$e_n(f)$, $i_n(f)$, spectral noise density.
 $e_n^2(f)$, $i_n^2(f)$, spectral noise power

white: $e_n(f) = e_{nw}$

$$e_n^2(f) = e_{nw}^2$$

e_n : voltage noise
 i_n : current noise.

$1/f$ noise:
 (pink) $e_n(f) = K_v/\sqrt{f}$

$$e_n^2(f) = K_v^2/f$$

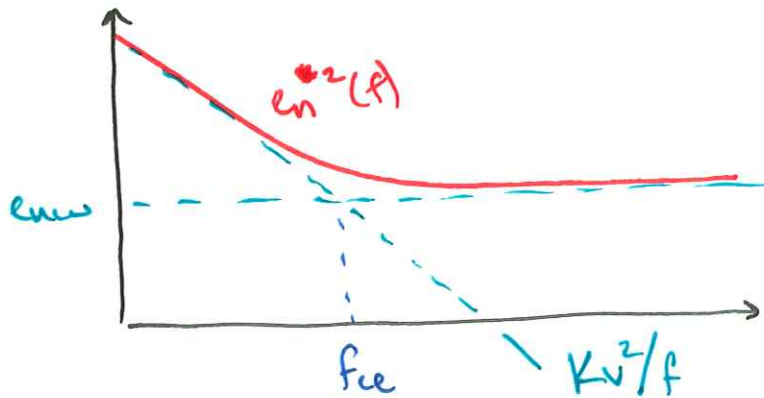
→ Always combine noise via squares (energy always adds)

$$e_{eq}^2 = e_{n1}^2 + e_{n2}^2 + e_{n3}^2 \dots$$

Op Amps present a combination of white & pink;

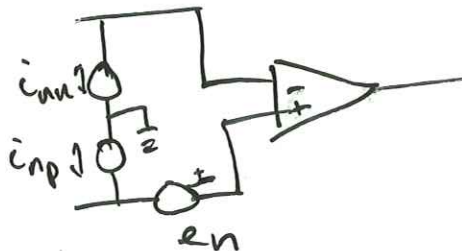
$$e_n^2 = e_{nw}^2 + K_v^2/f = e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right)$$

freq noise ~~corner~~ corner



← similar for i_{nn} & i_{np} as well

$$i_{nn}^2 = i_{nw}^2 + \frac{K_i^2}{f} = i_{nw}^2 \left(\frac{f_{ci}}{f} + 1 \right)$$



Total noise power:

$$E_n^2 = \int_{f_L}^{f_H} e_n^2(f) df$$

if white, f_L can go to 0
 $1/f$, f_L need lower limit.

$E_n = \sqrt{E_n^2} \rightarrow$ RMS value of noise

$I_n = \sqrt{I_n^2} \rightarrow$

$$SNR = 20 \log_{10} \left(\frac{V_s}{E_n} \right)$$

V_s = signal voltage IN RMS.

white: $E_n^2 = e_{nw}^2 (f_H - f_L)$
 $1/f$: $E_n^2 = K_v^2 \ln(f_H/f_L)$

f_L typically $\ll f_H$
 such that white $E_n^2 \approx e_{nw}^2 f_H$

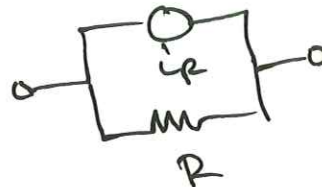
Component (Resistor) Noise

Thermal noise & flicker $\rightarrow$ typically Thermal dominant.

Thermal

$$e_R = \sqrt{4kTR}$$

$$i_R = \sqrt{4kT/R}$$



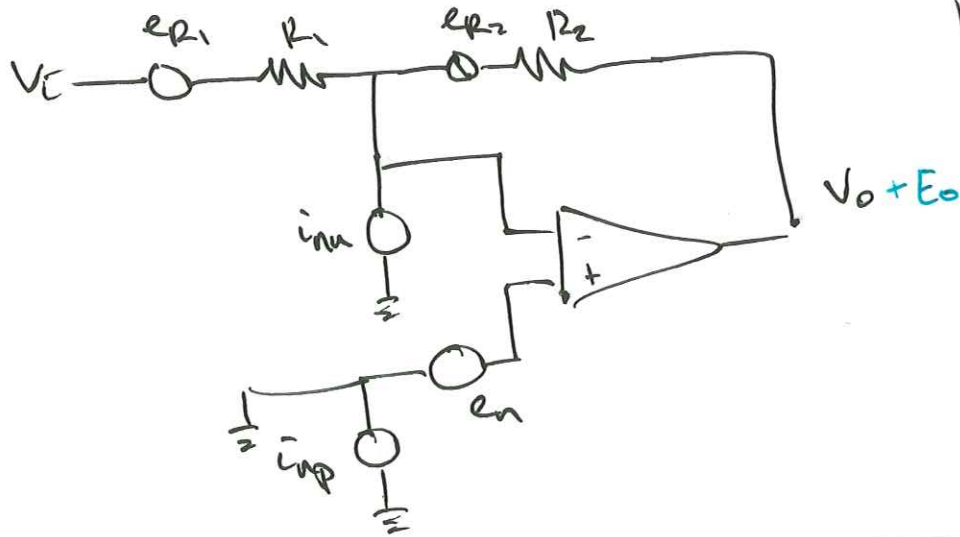
k = Boltzmann's constant

T = temp in K

Resistor networks: R_{eq}

$$e_{Req} = \sqrt{4kTR_{eq}}$$

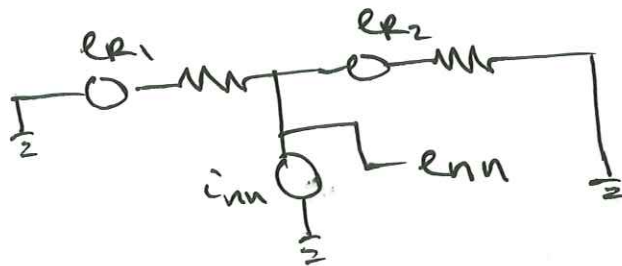
$$A = -\frac{R_2}{R_1}$$



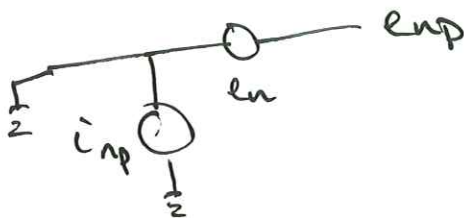
By superposition, assemble

$$E_o^2 \text{ or } e_{no}^2$$

OR use shortcut to find e_{ni}^2



$$e_{ni}^2 = e_{nn}^2 + e_{np}^2$$



NEB

look at Hw for integration of e_{no}^2

